

AgroStar: Small farms in India getting big help from the cloud

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AgroStar launched a multilingual mobile app using Google Cloud helping to boost productivity and increase income for small farmers in India.

[About AgroStar](#)



About AgroStar

Launched as an on-premises ecommerce platform selling farm supplies in 2008, AgroStar turned to Google Cloud to expand its offering.

Today one million small farmers in India use its mobile app to connect to a full-service platform that combines agronomy, data science, machine learning, and analytics to boost yields and improve income.

Industries: Agriculture, Forestry & Fishing

Google Cloud results

- Enables technology modules to speed loan processing,

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Agri-l
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identify crop diseases, and enhance supply chain logistics

- Delivers high rates of analytics processing to serve more customers while improving response times by 85%
- Streamlines deployments with no downtime to free time and resources for new verticals
- Supports knowledge center updates for 1 million mobile devices in near real time by implementing Firebase

week Goog

Location: India

Kubernetes

Google Cloud (<https://cloud.google.com/>)

Cloud

Pub/Sub (<https://cloud.google.com/pubsub/>),

Compute Engine

(<https://cloud.google.com/compute/>)

BigQuery (<https://cloud.google.com/bigquery/>).

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>).

Kubernetes (<https://cloud.google.com/kubernetes/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

IAM (<https://cloud.google.com/iam/>),

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

enterprise/

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud AI Platform (<https://cloud.google.com/ai-platform/>)

Cloud

Cloud SQL (<https://cloud.google.com/sql/>)

Operations

(<https://cloud.google.com/products/operations>)

Cloud Natural Language API

Cloud Storage (<https://cloud.google.com/storage/>)

Vision AI (<https://cloud.google.com/vision/>),

Cloud IAM (<https://cloud.google.com/iam/>)

Dialogflow Enterprise Edition

(<https://cloud.google.com/dialogflow-enterprise/>)

AI Platform (<https://cloud.google.com/ai-platform/>)

Tell us your challenge. We'd love to help

Cloud Video Intelligence API

(<https://cloud.google.com/video-intelligence/>)

Contact us (</contact>)

TensorFlow (<https://www.tensorflow.org/>)

Cloud Natural Language API

(<https://cloud.google.com/natural-language/>)

Vision AI (<https://cloud.google.com/vision/>)

An app to boost crop raise income

Dataflow (<https://cloud.google.com/dataflow/>)

Firebase (<https://firebase.google.com/>)

AgroStar (<https://corporate.agrostar.in/>) has launched a cloud-based mobile app that is helping to boost crop yields and encourage best practices for small farmers in India. Launched as an on-premises ecommerce platform selling farm tools in 2008, the firm turned to Google Cloud (<https://cloud.google.com/>) to expand its offering. It now uses cloud-based analytics and is deploying ML models to provide timely advice in five languages on everything from seed optimization, crop rotation, and soil nutrition to pest control.

"Today, just by clicking through our Android application, farmers learn about new, effective farming practices and receive advice customized to their crop and soil."

—Pritesh Gudge, Software
Engineer, AgroStar

A 2018 survey.

(<https://www.downtoearth.org.in/news/indias-deepening-farm-crisis-76-farmers-want-to-give-up-farming-shows-study-43728>)

underscored the demand for agricultural planning for Indian farmers. While farming remains a dominant sector in India, employing half of its labor force, 70 percent of small farmers – those cultivating fewer than three acres – said their crops are damaged by unforeseen weather and pests. An even higher number – 74 percent – say they lack access to farming-related information.

Widening that gap is the relative lack of access to new, higher yield seeds and improved soil analyses for small farmers, who must otherwise rely on traditional methods. "It could take a few years for innovative information to trickle down from universities to small, grassroots farmers," says Pritesh Gudge, AgroStar Software Engineer. "Today, just by clicking through our Android application, farmers learn about new, effective farming practices and receive advice customized to their crop and soil."

Connecting a million farmers in the cloud

Operating in the Indian states of Gujarat, Maharashtra, Rajasthan, Orissa, Bihar, and

Karnataka, AgroStar is closing the knowledge gap with a full-service, cloud-based SaaS solution – the only one of its kind in India. It combines agronomy, data science, and analytics to help farmers by providing a variety of resources.

AgroStar has reached over a million farmers through its Android app, the AgroStar Agri-Doctor. The mobile client is available as a web-based or full-featured native app. Both provide access to the firm's knowledge base hosted on Google Cloud, a Q&A forum that connects farmers to each other to help understand and better solve problems and to learn about innovative practices and products. Farmers can also click through to follow local and national market trends that help forecast crop prices.

In addition to the self-service knowledge base, AgroStar provides access to agronomy experts who use cloud-based analytics tools and historical data to provide season-and locale-specific advice to each farmer. "We are now tracking thousands of calls in 5 languages each day," says Pritesh.

The AgroStar app also provides links to purchase and then track the delivery of farm tools and supplies such as cultivators and fertilizers. An in-house platform manages fulfillment centers and a doorstep delivery network simplifies the supply chain while giving farmers what they need, when they need it. By procuring directly from the manufacturers and primary distributors of farm supplies, Agrostar is achieving cost savings, which it passes on to farmers.

Build fast, pivot faster

From the start, the human and environmental variables of farming in India, not to mention the volume of AgroStar's few hundred thousand monthly active users, made a highly scalable cloud-based solution inevitable. Farmers rely on the firm's Agri-Doctor app to provide advice in multiple languages on topics that range widely throughout three growing seasons, each with distinct crop nutrition and rotation cycles and farm implementation requirements.

"For farmers, the focus keeps changing every month, and every season," says Pritesh. "To serve our growing community, we needed a platform that could process images at high volume, fulfill tools and seed orders across thousands of miles, and respond to multilingual queries. We quickly moved away from spreadsheets and server-based solutions – we needed to build fast and pivot faster."

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—Pritesh Gudge, Software Engineer, AgroStar

Ending late-night deployments

The firm's first cloud experience was with an AWS solution. At the time, AWS was the only cloud provider in India, but AgroStar wanted to find a solution that was easier to use and offered better integration with Android devices. "Deployment and processing costs were very high, and the developer tools and documentation were not as intuitive as we needed," says Pritesh.

When Google Cloud service arrived in India in October 2017, AgroStar embarked on a platform re-implementation that made possible dramatic changes in the way it developed and deployed its solution. Using [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine/) (<https://cloud.google.com/kubernetes-engine/>) (GKE) for crop advice management and [Compute Engine](https://cloud.google.com/compute/) (<https://cloud.google.com/compute/>) for its production

application services, the firm built the backend for the Agri-Doctor discussion forum in only three weeks. The platform's microservice architecture is implemented in Python and Golang and deployed on Google Cloud.

AgroStar began to realize significant efficiencies in its build, deploy, and test cycles. "We previously needed to work overnight to deploy to production," says Pritesh. "Now using Google for Kubernetes (<https://cloud.google.com/kubernetes/>) containers and a rolling update strategy, we can deploy during the day without any problems or interruptions to service."

The move to Google Cloud streamlined AgroStar's stack. "We were running 12 independent instances on AWS," says Pritesh. "With Google Kubernetes Engine, we are deployed on a single cluster at a cost savings of \$1,300 per month and growing."

Improving customer response times by 85 percent

With a managed deployment capability, AgroStar can devote more time and resources to executing on its platform and Agri-Doctor app development plan. A strategic goal was managing customer response times as the firm grew its base. Google Cloud has helped the firm meet that goal, achieving an 85 percent improvement in customer response times even as traffic grew significantly.

"With our on-premises solution, we could handle around 100 customers daily, which took 30 to 50 minutes for each customer," says Pritesh. "We now

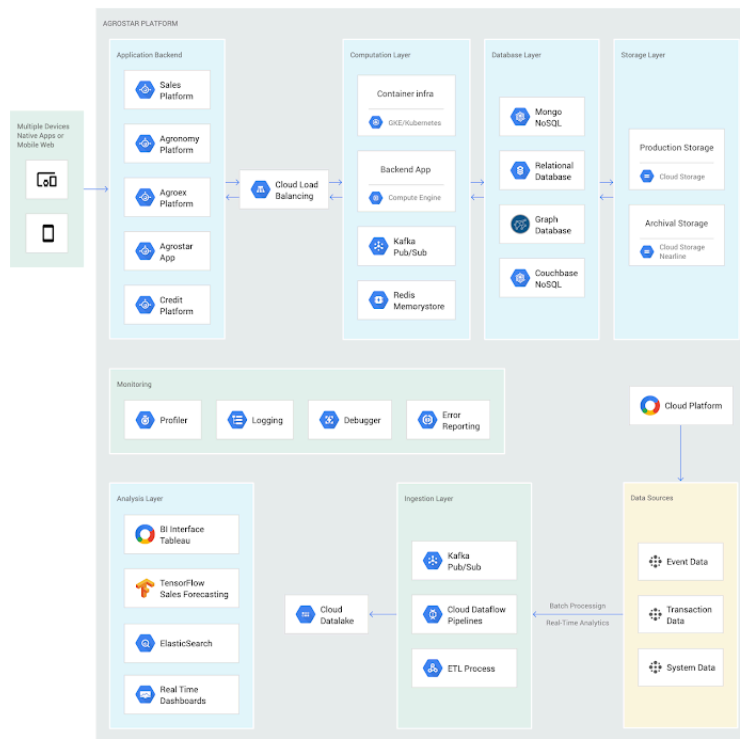
handle thousands of customers daily, taking only 4 to 5 minutes for each one."

AgroStar used [Firebase](https://firebase.google.com/) to implement its Agri-Doctor app. A real-time cloud database, Firebase provides an API that enables the Agri-Doctor advice forum to be synchronized across all its far-flung mobile clients, effectively sharing knowledge base updates with one million users in near real time.

Using cloud tools to manage and monitor

[Cloud Pub/Sub](https://cloud.google.com/pubsub/), [Kafka](https://cloud.google.com/pubsub/), and [Cloud Dataflow](https://cloud.google.com/dataflow/) manage data ingestion and queueing of event and transaction data to the analytics layer. [BigQuery](https://cloud.google.com/bigquery/) fetches and persists data to [Cloud Storage](https://cloud.google.com/storage/). [Cloud SQL](https://cloud.google.com/sql/) and dashboards powered by [Tableau](https://cloud.google.com/blog/products/gcp/best-practices-for-tableau-server-on-google-compute-engine) deliver farmer crop and soil profiles within minutes.

[Cloud IAM](https://cloud.google.com/iam/) helps AgroStar control access to all its cloud resources. And [Stackdriver](https://cloud.google.com/stackdriver/), the integrated logging aggregation capability for Google Cloud, helps monitor and speed debugging on every tier of the AgroStar solution.



AgroStar uses Golang and Python microservices and a variety of datastores (MongoDB, MySQL, Couchbase, Neo4j, and Elasticsearch) that are optimized to support concurrent usage in an interactive environment. Production applications are hosted on Google Cloud with Cloud Load Balancing to scale GKE and Compute Engine deployment and processing. Cloud Pub/Sub, Kafka, and Cloud Dataflow pipelines manage data ingestion and queueing of event and transaction data to the analytics layer, which uses BigQuery. Custom-built Cloud SQL and Tableau dashboards track strategic and tactical business metrics. A variety of ML models in TensorFlow are in the development phase.

Machine learning to enhance yields

AgroStar is developing a variety of ML components to improve responsiveness and extend its platform offerings.

To speed up the diagnosis of and treatment for crop blight, AgroStar is building a deep learning pipeline using TensorFlow

(<https://opensource.google.com/projects/tensorflow>). The pipeline relies on GoogLeNet

(https://www.youtube.com/watch?v=_XF7N6rp9Jw) models that use multi-layered convolutional visual pattern recognition. It will assess uploaded images to support a disease-detection capability on the mobile app. Based on the commercially successful AI algorithms that automated postal code processing, GoogLeNet offers improved performance and computational efficiencies by using a creative layering technique that distinguishes them from older, sequential recognition engines.

To improve its customer search experience, AgroStar is developing an ML pipeline that shrinks fetch times by suggesting tags mapped to stored data. Processed using TPUs, Cloud Natural Language (<https://cloud.google.com/natural-language/>) and Video AI

(<https://cloud.google.com/video-intelligence/>), the tags provide a metadata layer that supports queries in any of the ten natural languages that AgroStar farmers can use.

The AgroStar search pipeline consists of Long Short-Term Memory (LSTM) models of Recurrent Neural Networks. Recurrent networks exhibit "memory" through iterative processing and are distinguished from feedforward networks by a feedback loop connected to their past decisions,

ingesting their own outputs moment after moment as input.

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—Pritesh Gudge, Software Engineer, AgroStar

Implementing a recommendation engine

The firm is also adapting the Random Forests TensorFlow AI model to develop a crop and product recommendation engine. The model is trained by consuming numerical (rainfall, humidity, water availability per acre) and categorical (soil type, water sources) parameters to suggest appropriate products by season, region, and locale.

To simplify the product suggestion experience, AgroStar developers are testing Cloud Dialogflow (<https://cloud.google.com/dialogflow-enterprise/>), the Google Cloud conversational interface, to build a chatbot capability into its mobile app. The bot will track a farmer's crop schedules and answer simple questions by linking to the recommendation engine.

AgroStar is also extending its analytics platform with AI-powered sales planning and forecasting. Using linear regression models implemented in TensorFlow and powered by Cloud ML Engine (<https://cloud.google.com/ml-engine/>), the capability will enhance supply chain logistics as the company scales its operations across India.

To provide a credit on-demand offering for a range of seed-to-harvest cycle products, AgroStar is attempting to use Vision API (<https://cloud.google.com/vision/>) to create an AI model that will convert uploaded photos of customer application records into standard data formats. The firm's credit policy features a grace period in which farmers begin paying back loans after harvested crops go to market.

A versatile and friendly development ecosystem

AgroStar credits the convivial tools and documentation that Google Cloud offers and its incremental, pay-as-you-go pricing model for both the firm's success and its ability to manage growth.

"What Google Cloud offers is extremely good documentation and extremely simple-to-use tools and interfaces across all services," says Pritesh. "It helped us initially deploy our platform and at every scale that we have required since then, and its cost effectiveness enabled us to staff up to meet new feature milestones."

Contributors to this story

Pritesh Gudge: AgroStar Software Engineer. Pritesh was formerly a founder of Bauersafe, a farm protection solutions startup, and First Fit, a startup that focuses on mobility and functional techniques to promote general fitness and recovery. Pritesh earned a B.S. in Computer Science and Engineering at the Birla Institute of Technology and Science as well as degrees in Robotics and Deep Learning from Udacity.

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